

Daniel Park

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EDUCATION

Western University

Sept. 2021 – Apr. 2026

Specialization in **Computer Science**

- Member of **Western AI**, **Western CS Undergraduate Society** and Events Assistant at Western Sneaker Club.

Relevant Courses: Machine Learning, Databases, Statistical Programming, Unstructured Data, Operating Systems, Data Structures and Algorithms, Computer Networks.

PROFESSIONAL EXPERIENCES

Data & Machine Learning Intern

Sept. 2025 – Apr. 2026

HEAL

- Architected a cloud-native **ETL pipeline** on **AWS** using **Apache Airflow** to orchestrate **Google Places API** ingestion, detecting spatial anomalies for downstream **machine learning (ML)**.
- Engineered **data transformation** workflows in **Python (pandas, NumPy, regex)** to sanitize unstructured text, reducing anomalies and streamlining analytics preparation.
- Designed a scalable **data model** and storage architecture on **AWS** with rigorous schema validation, establishing governance protocols for high availability and pipeline reliability.
- Deployed predictive **machine learning models** with **Scikit-Learn** and **XGBoost** (clustering, time-series forecasting) to forecast retail density trends, translating complex data into visualizations for **strategic, data-driven decisions**.
- Implemented **MLOps** practices for continuous model validation, collaborating with **cross-functional stakeholders** to translate analytical outputs into clear business requirements.

Data Science Intern

Sept. 2024 – Aug. 2025

interRAI

- Managed the **ETL** lifecycle for high-volume clinical datasets using **PySpark**, optimizing distributed validation to ensure high-fidelity inputs for health research.
- **Modernized** legacy data pipelines into automated workloads utilizing **Spark SQL**, accelerating execution runtimes by **30%** and reducing manual processing overhead.
- Documented core **Spark** architectures and led technical training, accelerating onboarding and enabling unified **cross-functional collaboration**.

Front End Developer Intern

May 2024 – Aug. 2024

BERL

- Refactored legacy **React** applications and **Cascade CMS** templates utilizing **CSS Flexbox/Grid**, reducing system latency and streamlining digital content delivery.
- Transformed administrative interfaces into efficient workflows, enforcing accessibility standards and driving **continuous improvement** through rigorous peer code reviews.

Software Developer Intern

Sept 2023 – Apr. 2024

Cultureplex

- Engineered **Python (pandas, NumPy)** and **SQL** analytical frameworks to process complex datasets, accelerating large-scale data extraction for research initiatives.
- Optimized **PostgreSQL** databases via advanced **data modeling**, schema normalization, and query tuning, significantly reducing retrieval latency.
- Built a novel **ETL pipeline** leveraging **Python** and **LLMs (Generative AI)** to **transform** unstructured text into structured formats for **cross-functional** teams.
- Generated actionable visualizations (**Matplotlib/Seaborn**) to translate aggregated data into unified **Excel/CSV** structures, accelerating academic publications.

PROJECTS | [View All](#)

Canadian Ski Resorts Powder Predictor | *Airflow, PySpark, AWS (S3/RDS), XGBoost, Flask, React*

- Architected an end-to-end **MLOps pipeline** using **Apache Airflow** to automate daily ingestion of 7-day weather forecasts for 18 Canadian ski resorts via **Open-Meteo REST APIs**.
- Built a cost-effective **data lake on AWS S3** for raw storage and implemented a **PySpark ETL** job to compute complex meteorological features, including 3-day rolling snowfall and freeze-thaw cycles.
- Trained an **XGBoost Regressor** on 62,000+ historical records to predict a "Powder Score" (1-10), achieving an MAE of 0.01 and deploying model inference to an **AWS RDS PostgreSQL** warehouse.
- Developed a **Flask REST API** to serve insights to a dark-mode **React dashboard**, enabling users to filter and identify optimal skiing conditions.

TECHNICAL SKILLS & CERTIFICATIONS

Data Science & ML: Python, R, PyTorch, TensorFlow, Scikit-Learn, XGBoost, LLMs / Generative AI, Pandas, NumPy, Polars, SciPy, PySpark, Matplotlib, Seaborn, SHAP, UMAP, Yellowbrick, Jupyter

Languages & Web: SQL, Java, C, C++, C#, Bash, PHP, HTML/CSS, XML, Django, .NET, React

Cloud, Data Engineering & DevOps: ETL / ELT, Data Pipelines, Data Modeling, Airflow, AWS, Azure, GCP, Docker, Git, Maven, Elasticsearch, Lucene, MapReduce, Regex, Excel, CSV, Claude Code

Databases: MySQL, DynamoDB, Snowflake, Hadoop, Firebase, MongoDB, SQLite, PostgreSQL, MSSQL

- **AWS** Machine Learning Engineer Certificate - In Progress
- **Meta** Backend Developer Specialization Certificate